

expanded our product range from aluminium electrolytic capacitors to film capacitors and then small lithium titanate rechargeable batteries known as SLBs.

Innovating Solutions For India

Our growth strategy involves developing custom capacitor lineups to meet the specific needs of Indian applications such as EVs and smart electronics. By understanding the local demands, products are developed specifically to cater to them. For instance, capacitors for EVs must withstand high vibration and deliver high ripple current, while those for smart meters require reliable backup power under Indian specific temperature fluctuations.

India's EV and smart meter manufacturers face a critical challenge: conventional batteries can't handle 50°C temperatures and daily cycling, while traditional film capacitors have 2-3 year lead times that stall product launch. Our Film capacitors "EM series" address these problems, which I will discuss later in more detail.

Innovative Products For India

There are two product lines that I want to highlight here, Each one targets different market segments: (1) SLB series and (2) "EM series" Film capacitors.

SLB (Small Lithium Titanate Rechargeable batteries) are perfect for IoT devices, wearables, and energy harvesting applications because of their higher power density characteristics. The "EM Series" film capacitors, on the other hand, are designed especially to be used in EVs because of their compact size and high ripple current as compared to Aluminium Electrolytic Capacitors.

Small Lithium Titanate Rechargeable Batteries (SLBs)

Let me start with an example of a typical small electronics device that uses

our SLBs.: Samsung Galaxy S Series Pen. It uses our SLBs to power remote sensors inside the pen.

Unique Features:

- Long Life: They come with an exceptional life where they can maintain 80% capacity even after 25,000 charge/discharge cycles
- Ultra-Fast Charging: In just 3 minutes, it charges to approximately 80% capacity.
- Wide Temperature Range: It operates smoothly in a wider temperature range from -30°C to +60°C; high-temperature models are available up to +80°C, maintaining over 50% capacity even at temperature extremes!
- Low Internal Resistance: It enables efficient power delivery with minimal voltage drops. For part number SLB08115H100, internal resistance can be as low as ESR0.2Ω at 1kHz!

These batteries have passed UN38.3 transport standards and have shown extremely low risk of fire or smoke even when subjected to short circuits, crushing, or nail penetration tests. From a maintenance perspective, the life of 25,000-plus cycles means these batteries rarely need replacement, which cuts down long-term costs, especially for installation or repair in areas where servicing is difficult.



TIP: Scan this QR code for additional information on our SLB series batteries.

Basically, if fully charged, it can provide backup power for up to three to four minutes during discharge, while the other two alternative technologies like supercapacitor or lithium ion capacitor, last only a few seconds to a maximum of a minute. Overall, for the same physical size, the capacity and energy density of our SLB is much higher.

"EM Series" Film Capacitors

As mentioned earlier, the EV segment is growing rapidly in India. And, this segment needs new and innovative solutions. As a capacitor manufacturer, even though we are into passives, we are continuously developing new, innovative, advanced products.



Small Lithium Titanate Rechargeable Batteries (SLBs)

What we can see in the trend of EV requirements is that they want very high capacity, high ripple current, but in a very compact case size. Our "EM Series" film capacitors solve this very problem!

Unique Features:

- Flexible & Modular: Capacitors of this series offer a modular design so that multiple units can be connected in series or parallel configuration to help the customer achieve their required capacitance and ripple current.
- Compact Size: Our "EM Series" film capacitors are compact and cost-effective, yet they perform like the bulky capacitors. They can deliver 1,000V or more high-voltage performance in smaller footprints, and at lower costs than bulky film capacitors!
- Meet Automotive Standards: They are AEC-Q200 automotive standard qualified, which means that they can handle high vibration and a wide temperature range.



TIP: Scan this code for additional information about the "EM Series."

We are planning to start mass production of these innovative capacitors from early 2026. Samples are already available. Currently, we are supplying samples to a few customers for testing, but for mass production quantities, you need to wait until 2026.



TIP: Scan this Code to watch the video of EFY's discussion with Mr Ayumu Miyoshi.

Developed by Electronics For You team based on discussion with Mr Ayumu Miyoshi, Managing Director, Nichicon Electronics (India) Pvt Ltd

"EM Series" Film Capacitors

